The Axioms of Quantum Mechanics and the Arithmetic Structure of the Bost–Connes System

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Abstract

Starting from the finite p-adic reduction of the Bost–Connes system and the established dynamics of matter, we give a systematic account of five axioms: quantum states, composite systems, unitary evolution, observables, and Born probabilities. Complex amplitude patterns on a finite arithmetic space, together with its normalized inner product, support pure-state and density-operator descriptions. Joint configurations of the participating degrees of freedom realize the tensor structure, while self-adjoint exchanges that balance energy and group charge generate joint unitary evolution. The full joint state retains coherence among matter, background, and auxiliaries; local mixtures arise by reducing that same state. Spectral couplings correlate the components distinguished by an observable with orthogonal records. Statistical consistency determines their Born weights, while the same instrument specifies conditional states and successive event histories. Character-based matter patterns, persistent phases, and arithmetic spectral responses give these structures their physical content. The five axioms thus connect within a common process of states, interactions, and records, providing a unified description of the evolution of matter and its observable manifestations in this framework.

Keywords: Bost–Connes system; axioms of quantum mechanics; arithmetic state space; joint unitary evolution; quantum measurement; Born rule

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1 Introduction

Physical systems change state through interaction, and measuring devices form records by participating in that process. Records preserve the outcomes a device can distinguish, while correlations between the system and the device affect subsequent evolution. How states change, which distinctions are observable, and what statistical regularities records exhibit therefore call for a common physical description.

The Hilbert-space formulation of quantum mechanics organizes states, evolution, and observation into interrelated mathematical structures.[1] We organize this formulation around five axioms: quantum states, composite systems, unitary evolution, observables, and Born probabilities. Complex amplitudes describe states, joint spaces accommodate the participants in an interaction, and evolution and readout connect states to observable outcomes. Starting from the Bost–Connes (BC) framework, we examine the physical basis of these structures in the established processes of matter and observation, and how they consistently describe the same process.

The Kraus operator-sum representation describes local state changes arising from system–environment interactions.[2] Quantum instruments associate each outcome with its probability and the corresponding change of state.[3] Structural arguments for Born weights examine, respectively, additivity of projection probabilities, consistent assignments to positive effects, and symmetries under auxiliary transformations of entangled states.[4,5,6] These studies reveal mathematical connections among evolution, readout, and statistics. In the arithmetic framework considered here, the corresponding structures are realized by the matter carriers, exchanges, and recording processes already constructed.

Bost and Connes constructed a quantum statistical system with arithmetic symmetries from a number-theoretic Hecke algebra. Its partition function is the Riemann zeta function, and it exhibits a phase transition with spontaneous symmetry breaking.[7] In the finite p-adic reduction, the state-space dynamics constructs backgrounds and excitations and establishes joint cycles of exchange, retrieval, and restoration. It also constructs event instruments from internal records and develops the corresponding probability argument.[8] The unified theory of matter identifies stable amplitude and phase patterns as matter and obtains persistent phases and interaction responses through formation, maintenance, and propagation.[9] The cosmological account further follows matter, background, resources, and earlier records throughout the joint history, examining the information retained by local descriptions and their scope.[10] These constructions of state spaces, interactions, and records provide our common starting point.

In this framework, matter is described by complex amplitude patterns in a finite arithmetic state space. Relative amplitudes and phases specify a pattern’s structure; the background, thermal

environment, and return auxiliaries participate in its formation and maintenance. Exchanges change the states of the individual parts and establish correlations among them. Through actual spectral couplings, a measuring device correlates the structure it reads with orthogonal records. The changes in matter and the corresponding observational outcomes can therefore be calculated along the same joint state.

The relation between the whole and its parts is central to this process. Systems may have identical local populations yet differ in joint coherence; a local state can also become mixed while the full joint state retains coherence. Degrees of freedom left unresolved by one readout remain part of the whole, and their correlations can affect later interactions. Event statistics likewise rests on the complete output: dynamics determines the amplitude of each record component, and statistical consistency gives its squared-norm weight a probabilistic interpretation.

2 The Quantum-State Axiom

2.1 Mathematical Formulation of Quantum States

The quantum-state axiom specifies how physical states are represented in a complex Hilbert space: a pure state is represented by the ray determined by a nonzero vector, with vectors differing by a nonzero complex factor representing the same pure state; a general state is represented by a positive density operator of unit trace.[1]

Let $\mathcal{H}$ be the system's state space. For a normalized vector ψ, pure and general states are written, respectively, as

$$P_\psi = |\psi\rangle\langle\psi|, \qquad \rho \geq 0, \qquad \mathrm{Tr}\rho = 1. \tag{2.1}$$

Here a pure-state operator has rank one and satisfies $P_\psi^2 = P_\psi$. The normalized representatives ψ and $e^{i\theta}\psi$ give the same P_ψ, while relative phases remain in the cross terms between different components. The Hilbert space also specifies linear superposition, the inner product, and orthogonality, allowing states to be represented in different orthonormal bases.

In the present framework, finite arithmetic patterns and the operations acting on them realize these structures. Pure and general states can be described within the same state space and its positive structure.

2.2 Construction of the Arithmetic State Space

At a given finite resolution, a physical pattern is described by a set of complex amplitudes. Following the finite-reduction construction, fix an odd prime p and a depth $K \geq 1$, and write

$$\begin{aligned} X_{p,K} &= \mathbb{Z}/p^K\mathbb{Z}, \\ d &= p^K, \\ \mathcal{H} &= L^2(X_{p,K}, \mathbb{C}). \end{aligned} \tag{2.2}$$

Each residue class corresponds to an arithmetic address, and the function $f(x)$ gives the complex amplitude at that address. Amplitude ratios and phase differences between addresses jointly

determine the pattern. Here $X_{p,K}$ is the finite arithmetic base space; the position of propagating matter is described separately by the propagation construction.[8,9]

The state-space dynamics uses normalized counting Haar measure and has already established the inner product, point basis, and directional decomposition.[8] For $f, g \in \mathcal{H}$,

$$\begin{aligned} \langle f, g\rangle &= \frac{1}{d}\sum_{x \in X_{p,K}} \bar{f(x)} g(x), \\ |x\rangle &= \sqrt{d}\ \ \mathbf{1}_{\{x\}}. \end{aligned}$$

(2.3)

Thus $\langle x|y\rangle = \delta_{xy}$, and $\mathcal{H}$ has complex dimension d. Finite dimensionality guarantees completeness of this inner-product space. A pattern expands in the normalized point basis as

$$\begin{aligned} f &= \sum_{x \in X_{p,K}} c_x |x\rangle, \\ c_x &= \frac{f(x)}{\sqrt{d}}, \\ \| f \|^2 &= \sum_x |c_x|^2. \end{aligned}$$

(2.4)

The same measure fixes the normalization relating function values to expansion coefficients. The shell decomposition, phase operations, and joint exchanges all use this inner product.

This construction is also compatible with the positive structure of the finite phase algebra. For $a \in C(X_{p,K})$, the uniform positive normalized functional is

$$\begin{aligned} \varphi_0(a) &= \frac{1}{d}\sum_x a(x), \\ \langle f, g\rangle &= \varphi_0(f^* g). \end{aligned}$$

(2.5)

Here f^* denotes complex conjugation. A positive functional produces a Hilbert-space representation through the Gelfand–Naimark–Segal (GNS) construction.[11] Since $\varphi_0(f^* f) = 0$ holds only for $f = 0$, its GNS space is precisely the finite L^2 space above, on which the phase algebra acts by multiplication operators. The original complex amplitude space, normalized counting inner product, and positive algebraic structure are therefore consistent.

This linear space also accommodates coherent superpositions of patterns. For two patterns f, g, the coherent combination $\psi = af + bg$ remains in $\mathcal{H}$. The original phase operations, directional transformations, and exchange matrix elements act linearly, so their action on this combination is

determined jointly by its components.[8] Cross terms encode the coherence between the two components.

Normalization separates the overall scale of a nonzero pattern. Its pure-state representation is

$$P_{[f]} = \frac{|f\rangle\langle f|}{\langle f, f\rangle}, \qquad f \neq 0.$$

(2.6)

Multiplying f by any nonzero complex number produces the same squared-modulus factor in the numerator and denominator, leaving $P_{[f]}$ unchanged. Conversely, if two such rank-one projections coincide, their ranges are the same one-dimensional subspace, so the two patterns differ by a nonzero complex factor. Pure-state rays thus correspond exactly to normalized patterns modulo overall phase.

Relative phase remains present in the pure-state operator. For orthonormal f, g and $|a|^2 + |b|^2 = 1$,

$$P_\psi = |a|^2 P_f + |b|^2 P_g + a\bar{b}\,|f\rangle\langle g| + \bar{a}b\,|g\rangle\langle f|.$$

(2.7)

The last two terms vary with relative phase and distinguish a coherent superposition from a mixture with the same diagonal populations. An interaction connecting these components acts on these cross terms. Retaining only the squared moduli of the component amplitudes does not represent the full pattern.

The original cycle admits general preparations and retains their populations and coherence in a density operator.[8] For nonnegative preparation weights w_ℓ,

$$\begin{aligned} &\rho = \textstyle\sum_\ell w_\ell P_{\psi_\ell}, \\ &w_\ell \geq 0, \qquad \textstyle\sum_\ell w_\ell = 1. \end{aligned}$$

(2.8)

This operator is positive and has unit trace. Conversely, finite-dimensional spectral decomposition expresses every positive unit-trace operator in this form, so the same representation accommodates both pure and mixed states. A given ρ can have more than one pure-state ensemble decomposition.[12] As the current system state, ρ retains the matrix elements that participate in subsequent linear operations. If the preparation apparatus also retains records correlated with the system, the complete description includes those records.

2.3 Quantum Description of Matter Patterns

The identification of matter gives these quantum states concrete physical content. A complete decomposition into shell-average and directional components has been established in the finite state space.[8] For $0 \le j < K$, the valuation shell S_j consists of residue classes divisible by p^j but not by p^{j+1}; the terminal shell is $S_K = \{0\}$. The shells are pairwise disjoint and cover $X_{p,K}$. Their normalized shell-average states are

$$e_j = \frac{1}{\sqrt{|S_j|}} \sum_{x \in S_j} |x\rangle.$$

(2.9)

The shell-average states $e_0, \ldots, e_K$ span $\mathcal{C}$, and the nontrivial directions within the shells span $\mathcal{M}$. Denoting the corresponding orthogonal projections by Π and P, we have

$$\begin{aligned} &\mathcal{H} = \mathcal{C} \oplus \mathcal{M}, \\ &\Pi + P = I, \\ &\rho = \Pi\rho\Pi + P\rho P + \Pi\rho P + P\rho\Pi. \end{aligned}$$

(2.10)

The first two terms describe the populations and coherence within each subspace, while the last two retain coherence between them. The background reference and directional excitations can therefore be compared within a single system state, and formation exchanges proceed through its matrix elements. The original process specifies the background preparation used in a cycle. The constant-function reference, shell-average subspace, and actual background density operator each have their respective roles.

The unified theory of matter identifies stable patterns by pattern preservation.[9] For the original shell-preserving response $T_{\mathrm{el}}(s_*)$, a directional pattern $\psi \in \mathcal{M}$ satisfies

$$\begin{aligned} &T_{\mathrm{el}}(s_*)\psi = \lambda\psi, \\ &\lambda \neq 0, \\ &P_{[T_{\mathrm{el}}(s_*)\psi]} = P_{[\psi]}. \end{aligned}$$

(2.11)

Here s_* is a regular spectral point of the original response. The unified theory of matter has established the equivalence between a common nonzero response factor and invariance of the pure-state projection: relative amplitudes and phases across the pattern remain unchanged, and the response changes only a common factor. The pure-state ray thus corresponds directly to the

shape of a matter pattern, while its actual formation, maintenance, and exit are determined by the complete cycle.

Basic states provide a concrete realization of this relation. Let $G_{p,K} = (\mathbb{Z}/p^K\mathbb{Z})^\times$ be the unit group, of order $N = (p-1)p^{K-1}$, with natural reduction $\pi: G_{p,K} \to G_{p,1}$. For a nontrivial multiplicative character χ on the species layer, the unified theory of matter gives the directional pattern[9]

$$|\psi_\chi\rangle = \frac{1}{\sqrt{N}} \sum_{u \in G_{p,K}} \chi(\pi(u))|u\rangle.$$

(2.12)

At the selected spectral point, write the eigenvalue of the shell-preserving response along this direction as $R_\chi(s_*)$. When it is nonzero, the direction satisfies the pattern-preservation criterion (2.11) and constitutes a basic state. All unit addresses have the same amplitude modulus, while the character specifies their relative phases. Different directions can have identical diagonal populations in the point basis yet retain different matter patterns in their off-diagonal matrix elements. Identifying matter states therefore requires the complete quantum-state representation. Their physical masses and propagation are further determined by persistent phases and responses under the original interactions.[9]

Complex amplitudes and the normalized inner product of finite arithmetic patterns provide the Hilbert-space structure for quantum states. Overall phase leaves the pure-state ray of a normalized pattern unchanged, and density operators give a common representation of pure and mixed states. Pattern preservation connects this description to the identification of matter and its sustained dynamics. When matter participates in exchanges, the background, auxiliaries, and records, together with their correlations, must also enter the joint description.

3 The Composite-System Axiom

3.1 Tensor Structure of Composite Systems

Physical systems form a whole through interaction. For two separately specifiable sets of degrees of freedom A and B, the composite-system axiom describes their joint state in a tensor-product space: [1]

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B.$$

(3.1)

Independent pure preparations are represented by product vectors; a general joint pure state can be a linear combination of such vectors. Independent mixed preparations correspond to $\rho_A \otimes \rho_B$, and a general joint state is represented by a density operator Ω_{AB} on $\mathcal{H}_{AB}$. An operator acting only on A takes the form $A_0 \otimes I_B$, while the local state of A is $\mathrm{Tr}_B \Omega_{AB}$.

This axiom specifies both how systems are combined and how their joint information is retained. The participants in the original formation exchanges, coherent retrieval, and recording processes give the construction of the joint state space concrete physical content.

In the original cycle, the system space $\mathcal{H}_S$ contains both shell-average and directional components. The thermal environment $\mathcal{E}_T$ and return auxiliary $\mathcal{E}_R$ carry the states required for exchange and retrieval.[8] Including the other degrees of freedom that continue to participate, the full space is[10]

$$\mathcal{K} = \mathcal{H}_S \otimes \mathcal{E}_T \otimes \mathcal{E}_R \otimes \mathcal{E}_{\mathrm{rest}}.$$

(3.2)

Each factor corresponds to a set of participating degrees of freedom in the original process. Within the system, $\mathcal{C} \oplus \mathcal{M}$ describes orthogonal components of a single pattern; the tensor factors in (3.2) jointly specify the states of the system and its auxiliaries. These two forms of combination describe internal structure and system composition, respectively.

For two finite configuration sets X_A, X_B, a joint configuration is specified by an ordered pair (x, y). With the original normalized counting measures, product patterns are related by

$$J(f \otimes g)(x, y) = f(x)g(y),$$
$$\langle J(f \otimes g), J(f' \otimes g') \rangle$$
$$= \langle f, f' \rangle \langle g, g' \rangle.$$

(3.3)

The normalized point basis $|x\rangle \otimes |y\rangle$ corresponds one-to-one to the joint point basis $|(x, y)\rangle$, and the two bases span their respective spaces. The joint configurations of independent degrees of freedom and their amplitude inner products therefore give

$$L^2(X_A \times X_B) \cong L^2(X_A) \otimes L^2(X_B).$$

(3.4)

This correspondence preserves the individual operations: a linear operation acting only on x, independently of y, becomes $A_0 \otimes I_B$; an operation connecting both sets of states acts on the joint space. Orthogonal environment and record states enter the original cycle in the same way.

The arithmetic structure also yields factorizations of certain internal mode spaces. For example, the unified theory of matter uses a coprime decomposition of the unit group at the species layer to obtain a tensor representation of the $(p-1)$-dimensional species space.[9] This shows that a single system can contain mode factors on which operations act separately. The actual composition of a system and its auxiliaries is determined by the participants in (3.2). Both use the same tensor structure in their respective spaces.

3.2 Entanglement in Formation Exchanges

The tensor product accommodates independent preparations as well as correlations produced by exchange. The state-space dynamics has already constructed the joint formation interaction between the background and directional excitations.[8] Consider a shell-average state $|e_a\rangle$ and a directional state $|\nu\rangle$ of the system, together with two orthogonal thermal-environment states $|u\rangle, |v\rangle$. The system states are also orthogonal, and the joint endpoints $|e_a, v\rangle$ and $|\nu, u\rangle$ have matching total energies and total group charges. On this block, the original formation exchange gives

$$W_{a\nu}|e_a, v\rangle = \cos\alpha\, |e_a, v\rangle - \sin\alpha\, |\nu, u\rangle.$$

(3.5)

Here α is the original exchange angle, and (3.5) is written in the interaction picture. Formation of a direction from the background and the accompanying environmental change appear in the same joint output; the reverse process uses the other input of the same interaction.

Write the pure output of (3.5) as $|\Psi_\alpha\rangle$. When $\sin\alpha\cos\alpha \neq 0$, the two system components are correlated with two orthogonal environment components, and the coefficient matrix has rank two. The output therefore cannot be expressed as a product of one system vector and one environment vector. Such a nonproduct pure state is entangled.[13] The original formation exchange thus establishes a coherent connection between the system transition and the corresponding change in the environment.

The cross terms in the full joint state are

$$-\sin\alpha\cos\alpha\left(|e_a, v\rangle\langle\nu, u| + |\nu, u\rangle\langle e_a, v|\right).$$

(3.6)

These matrix elements connect the two joint endpoints. They can participate in the original retrieval and subsequent exchanges, and cannot be replaced by the separate populations of the system and environment. For the original thermal mixed preparation, all input components propagate with their preparation weights, and the joint density operator retains the complete output.

Correlations in general mixed states also require a distinction. A state expressible as

$$\Omega_{AB} = \sum_\ell w_\ell \rho_A^{(\ell)} \otimes \rho_B^{(\ell)},$$
$$w_\ell \geq 0, \quad \sum_\ell w_\ell = 1$$

(3.7)

is called separable and may still contain classical correlations between preparations; a joint state admitting no such decomposition is entangled.[14] The original formation output shows that an actual joint interaction in the framework can generate the latter from a product input. The joint space therefore describes independent preparations, correlated mixtures, and entanglement generated by coherent exchange.

3.3 Local Reduction of Joint States

A local state describes the properties of a selected system at a given time. The state-space dynamics has already obtained the reduction relation from the output of a complete cycle.[8] For a joint pure state, choose an orthonormal system basis and write

$$
\begin{aligned}
|\Psi\rangle &= \sum_i |i\rangle \otimes |\eta_i\rangle, \\
\rho_S &= \mathrm{Tr}_E |\Psi\rangle\langle\Psi|, \\
\langle i|\rho_S|j\rangle &= \langle \eta_j|\eta_i\rangle.
\end{aligned}
$$

(3.8)

The environment vectors η_i already include the amplitudes of the corresponding system components. Their squared norms give the diagonal populations, while their overlaps determine the cross terms retained by the system. Summing linearly over the weighted pure components of a mixed joint state gives the same reduction formula.

For the original formation output (3.5), the environment states u, v are orthogonal, so

$$
\begin{aligned}
\rho_S &= \cos^2\alpha\, P_{e_a} + \sin^2\alpha\, P_\nu, \\
\mathrm{Tr}\rho_S^2 &= \cos^4\alpha + \sin^4\alpha.
\end{aligned}
$$

(3.9)

When both components are nonzero, the local system state is mixed while the joint state remains pure. The cross terms absent from the system state remain in the system–environment correlations of (3.6). The partial trace restricts the description to the system without deleting these correlations from the whole.

Figure 3-1 compares the joint and local descriptions of the same exchange output, showing where the coherence terms reside in each description.

Output of the original formation exchange
$|\Psi_\alpha\rangle = c|e_a, v\rangle - s|\nu, u\rangle$

Joint state	Partial trace over the environment	**Reduced system state**
$\Omega_{SE} = \begin{pmatrix} c^2 & -cs \\ -cs & s^2 \end{pmatrix}$	- - - - - →	$\rho_S = \begin{pmatrix} c^2 & 0 \\ 0 & s^2 \end{pmatrix}$
Cross terms retained on the joint support		Orthogonal environment components
Purity: 1		No local cross terms; purity: $c^4 + s^4$

Figure 3-1. Joint coherence in a formation exchange and its local reduction. Let $c = \cos\alpha$ and $s = \sin\alpha$. The joint-state matrix is written on the two-dimensional support $\{|e_a, v\rangle, |\nu, u\rangle\}$, and the reduced-state matrix in the system basis $\{|e_a\rangle, |\nu\rangle\}$. Orthogonality of the environment states u, v removes the cross terms from the system state under the partial trace; the joint state retains this coherence. The dashed arrow denotes a reduction at the same instant.

The correspondence between local states and local operations is given by

$$\mathrm{Tr}_S(\rho_S A) = \mathrm{Tr}_{SE}\left[\,\Omega_{SE}(A \otimes I_E)\,\right]$$

(3.10)

This identity shows that ρ_S reproduces every current linear readout acting only on the system. A later interaction involving both the system and environment must be evaluated on their joint state. Identical marginals do not guarantee identical subsequent joint states.

The cosmological account provides a concrete example of this informational distinction in the full BC system.[10] In the standard integer basis ε_m, take distinct primes p, q and set

$$\begin{aligned}\Psi_\pm &= \frac{\varepsilon_1 \pm \varepsilon_{pq}}{\sqrt{2}},\\ \omega_\pm(A) &= \langle \Psi_\pm, A\Psi_\pm \rangle.\end{aligned}$$

(3.11)

Let $\mathfrak{A}_r$ denote the observational channel algebra generated in the cosmological account by the phase operations, scaling, and adjoint scaling at prime r. The established result is that the two states have identical restrictions to every single channel, while joint scaling distinguishes them:

$$\begin{aligned}&\omega_+|_{\mathfrak{A}_r} = \omega_-|_{\mathfrak{A}_r} \quad \text{(all primes } r\text{)},\\ &\omega_\pm(O_{pq}) = \pm 1,\end{aligned}$$

(3.12)

Here $O_{pq} = \mu_p \mu_q + (\mu_p \mu_q)^*$, where μ_p, μ_q are the prime scalings of the parent system. The distinction lies in joint coherence and cannot be recovered from single-channel statistics. These prime-indexed observational algebras generally do not commute with one another; the tensor decomposition and partial trace for a system and its auxiliaries, by contrast, are determined by their joint configurations. Both descriptions retain the joint information accessible to the interactions under study.

Joint configurations and their amplitude inner product determine the tensor product. Formation exchanges establish correlations and entanglement between system and environment in that space. The partial trace gives a local state whose coherence is determined by overlaps of the environment components. Joint correlations omitted from the local description can still participate in later interactions, so the input required for further evolution depends on the actual participants. Retaining matter, background, auxiliaries, and records together allows their successors to be followed from the original joint state.

4 The Unitary-Evolution Axiom

4.1 Unitary Evolution of Closed Systems

The unitary-evolution axiom describes the continuous change of a closed system in time. On a given joint state space, evolution is represented by a unitary operator and hence preserves inner products, state normalization, and distinguishability between preparations.[1] If $U(t, t_0)$ denotes evolution from t_0 to t, then

$$\begin{aligned} &U(t, t_0)^\dagger U(t, t_0) = I, \\ &|\Psi(t)\rangle = U(t, t_0)|\Psi(t_0)\rangle, \\ &\Omega(t) = U(t, t_0)\Omega(t_0)U(t, t_0)^\dagger. \end{aligned}$$

(4.1)

A continuously differentiable unitary evolution has a self-adjoint generator. We retain $\hbar$ to make the dimensions of time and energy explicit. Define

$$\begin{aligned} &H(t) = i\hbar\,\partial_t U(t, t_0)U(t, t_0)^\dagger, \\ &H(t)^\dagger = H(t), \\ &i\hbar\,\partial_t|\Psi(t)\rangle = H(t)|\Psi(t)\rangle. \end{aligned}$$

(4.2)

Self-adjointness follows directly by differentiating $UU^\dagger = I$. The corresponding density operator satisfies

$$i\hbar\,\dot{\Omega}(t) = [H(t), \Omega(t)].$$

(4.3)

A fixed generator gives $U(t, t_0) = e^{-i(t - t_0)H/\hbar}$; time-dependent controls are described by time-ordered evolution on the same joint space. Both forms preserve the unitarity in (4.1).

In this framework, the scope of a closed description is set by the actual participants. During exchange with a thermal environment, matter can change its occupations, purity, and coherence. Including the environment, return auxiliaries, and controls in the joint state gives a complete description of the exchange. The unitary-evolution axiom is realized here as a reversible joint interaction among these participants.

4.2 Dynamics of Arithmetic Exchanges

Let Π_j be the orthogonal projection onto the entire jth valuation shell. The free generator is $H_K = \sum_{j=0}^{K} E_j \Pi_j$. In the common energy unit E_*, the original arithmetic energy levels are $E_j = E_* j \log p$. The shell projections and free evolution satisfy

$$\begin{aligned} &\Pi_j = \sum_{x \in S_j} |x\rangle\langle x|, \qquad 0 \le j \le K, \\ &U_f(t) = \sum_{j=0}^{K} e^{-itE_j/\hbar} \Pi_j, \\ &\Pi_j \rho(t) \Pi_\ell = e^{-it(E_j - E_\ell)/\hbar} \Pi_j \rho(0) \Pi_\ell. \end{aligned} \tag{4.4}$$

All directions within a shell acquire the same free phase, while coherence between shells evolves with their energy gaps. The original shell structure thus determines both the decomposition of free evolution and the frequencies of relative phases.[8]

Matter formation involves actual exchanges between different directions. The state-space dynamics has already constructed $F_{a\nu} = |e_a\rangle\langle\nu|$ from directional projections, character phases, and shell shifts. Combining this operator with an environment transition that balances energy and group charge gives

$$\begin{aligned} &L_{a\nu} = F_{a\nu} \otimes |v\rangle\langle u|, \\ &B_{a\nu} = L_{a\nu} - L_{a\nu}^{\dagger}, \\ &H_{a\nu} = i\hbar \kappa_{a\nu} B_{a\nu}. \end{aligned} \tag{4.5}$$

Here $\kappa_{a\nu}$ is the exchange frequency. The original balancing makes the total bare energies and total group charges of $|\nu, u\rangle$ and $|e_a, v\rangle$ equal, and the same self-adjoint $H_{a\nu}$ describes exchange in both directions. Since $B_{a\nu}^{\dagger} = -B_{a\nu}$, the interaction-picture exchange over duration $t_{a\nu}$ satisfies

$$\begin{aligned} &W_{a\nu}(t_{a\nu}) = e^{-it_{a\nu}H_{a\nu}/\hbar} = e^{\alpha B_{a\nu}}, \\ &\alpha = \kappa_{a\nu} t_{a\nu}. \end{aligned} \tag{4.6}$$

This is precisely the exchange that generates background–excitation entanglement in Chapter 3. Amplitude transfer, environmental backreaction, and the forward and reverse paths are contained in the same exponential evolution. The original arithmetic matrix elements and the compensating environmental transition determine the interaction, while the exchange angle relates coupling strength to elapsed physical time.

The complete cycle composes these interactions in their actual order. For the original thermal exchange, coherent retrieval, return, and restoration stages, the joint unitary can be written as

$$\begin{aligned}\mathbb{U}_{\mathrm{cyc}} &= e^{-i\tau H_0/\hbar} U_{\mathrm{res}} U_{\mathrm{back}} U_{\mathrm{out}} U_{\mathrm{th}},\\ H_0 &= H_K + H_R + H_T.\end{aligned}$$

(4.7)

Here H_R, H_T are the Hamiltonians governing the free evolution of the return auxiliary and thermal environment, respectively, and τ is the total elapsed time, including free intervals and all exchange stages; identity factors on the individual degrees of freedom are suppressed. Each exchange in the original construction commutes with the total bare energy, so free evolution can be collected into the leftmost factor while the interaction order is retained. Control degrees of freedom and their actions enter the joint description at the corresponding stages. Durations of serial stages add, whereas simultaneous arms share the same time interval.[8,10] Reconstructing the pattern at the end of a cycle amounts to representing this joint state in the point basis.

The persistent phase of matter is realized by the reflection-based two-arm construction in the unified theory of matter. The isometry $\mathcal{J}_s$ represents the prepared complete matter carrier, with projection $P_s = \mathcal{J}_s \mathcal{J}_s^\dagger$ and fixed preparation reflection $S_s = 2P_s - I$. The self-adjoint exchange X includes spectral access, address matching, and all auxiliaries and failure outputs. Two coherent arms of equal strength implement X and $S_s X S_s$ simultaneously, with their amplitudes combined in the same complete output space. They give the generator and its matter representation[9]

$$\begin{aligned}\widehat{H_X} &= \frac{g}{2}(X + S_s X S_s),\\ h_X &= g\mathcal{J}_s^\dagger X \mathcal{J}_s,\\ \widehat{H_X}\mathcal{J}_s &= \mathcal{J}_s h_X.\end{aligned}$$

(4.8)

Here g is expressed in units of energy. Cross terms between the range of the projection and its complement cancel in the sum of the two arms, so sustained evolution on the prepared carrier satisfies

$$e^{-it\widehat{H_X}/\hbar}\mathcal{J}_s = \mathcal{J}_s e^{-ith_X/\hbar}.$$

(4.9)

Thus h_X is the restriction of the full generator to the prepared matter carrier, and its eigenvalue differences determine the persistent phase. The energy described by h_X belongs to the active dynamics, whereas H_K describes the bare shell energy. Control energy supply and switching are

included in the full interaction, while the original thermal exchanges remain balanced in total bare energy.

From free phase evolution through formation exchange to the maintenance of matter, the unitary operations in the framework have specified participants and generators. Continuous evolution turns these actual matrix elements into changes of state in physical time.

4.3 Local Description of Joint Evolution

Unitary evolution of a joint state allows nonunitary changes in its local states. For an initial product preparation $\rho \otimes \eta$ of system and environment, the evolved system state is

$$\mathcal{E}(\rho) = \mathrm{Tr}_E\left[\, U(\rho \otimes \eta)U^{\dagger} \,\right].$$

(4.10)

This auxiliary realization exhibits the Stinespring dilation structure of completely positive maps.[15] Here the original cycle specifies the joint unitary U and environment preparation η. Taking $\eta = \sum_j \pi_j |j\rangle\langle j|$ and an orthonormal output basis $\{|\mu\rangle\}$ gives

$$\begin{aligned} K_{\mu j} &= \sqrt{\pi_j}\ \langle \mu | U | j \rangle, \\ \mathcal{E}(\rho) &= \sum_{\mu,\, j} K_{\mu j} \rho K_{\mu j}^{\dagger}, \\ \sum_{\mu,\, j} K_{\mu j}^{\dagger} K_{\mu j} &= I. \end{aligned}$$

(4.11)

Equation (4.11) is the Kraus representation of the local quantum channel. Its operator-sum structure agrees with the characterization of finite-dimensional completely positive maps, and the completeness relation guarantees trace preservation.[2,16] Complete positivity is also evident directly from the joint process: any spectator system can be included through an identity factor, and joint positivity is preserved by unitary evolution and the partial trace.

The formation output (3.5) and local state (3.9) illustrate this relation. Orthogonal environment components make the joint cross terms absent from the reduced system state, exhibiting the basic mechanism of environment-induced decoherence.[17] If the same environment participates in a later return, correlations retained in the joint state can still affect the system, and the calculation proceeds from the joint state after exchange.

The cosmological account expresses this continuous relation on the full state.[10] For a fixed system operator A,

$$\frac{d}{dt}\mathrm{Tr}(\rho_S A) = \frac{i}{\hbar}\mathrm{Tr}\big(\,\Omega[H, A \otimes I_E]\,\big).$$

(4.12)

The commutator on the right includes system–environment correlations, so subsequent changes are determined by the joint state. A fixed environment preparation and interaction give the same system channel when each fresh environment is independent of the existing whole. Reusing an old auxiliary instead requires continuing from the joint state that retains its correlations.

The original arithmetic matrix elements form self-adjoint generators through joint exchanges that balance energy and group charge, driving matter, background, and auxiliaries together. Joint unitaries preserve inner products while changing the populations and correlations of the parts; local thermalization and changes of coherence arise by reducing this process. A measuring device likewise participates through actual interactions in the same evolution, correlating distinctions in the state with records.

5 The Axiom of Observables

5.1 Spectral Representation of Observables

The axiom of observables represents ideal physical readouts by self-adjoint operators. In finite dimensions, a self-adjoint operator has real eigenvalues and an orthogonal spectral decomposition, connecting possible readouts to the state subspaces that distinguish them.[1] Let

$$\begin{aligned} A = A^\dagger = \sum_\kappa a_\kappa E_\kappa, \qquad & a_\kappa \in \mathbb{R}, \\ E_\kappa E_\lambda = \delta_{\kappa\lambda} E_\kappa, \qquad & \sum_\kappa E_\kappa = I. \end{aligned}$$

(5.1)

Each E_κ corresponds to a distinguishable spectral outcome, and a_κ calibrates its readout. A degenerate eigenvalue corresponds to an entire spectral subspace; individual vectors in that subspace need not be identified as different outcomes.

Self-adjointness also ensures that the pairing with a state is real. For a positive unit-trace ρ, this linear pairing is

$$\ell_\rho(A) = \mathrm{Tr}(\rho A) = \sum_\kappa a_\kappa \mathrm{Tr}(\rho E_\kappa).$$

(5.2)

Once record probabilities have been established, this pairing acquires the meaning of a statistical average. Realizing a spectral readout first requires an apparatus interaction that correlates the spectral components of A with distinguishable records.

5.2 Formation of Physical Readouts

The original arithmetic operations already generate a noncommutative local observable algebra.[8] For a nonzero conjugate address pair $D = \{a, -a\}$, write $P_{\pm a} = |\pm a\rangle\langle \pm a|$, and let $\iota|x\rangle = |-x\rangle$ be the linear address-negation operation. Point projections and negation give the matrix units

$$\begin{aligned} E_{++} &= P_a, \qquad E_{--} = P_{-a}, \\ E_{+-} &= P_a \iota P_{-a}, \qquad E_{-+} = P_{-a} \iota P_a. \end{aligned}$$

(5.3)

They generate $M_2(\mathbb{C})$ on $\mathcal{H}_D = \mathrm{span}\{|a\rangle, |-a\rangle\}$. Its self-adjoint part can be written as

$$A = a_0 I + \sum_{i=1}^{3} a_i \sigma_i, \qquad a_i \in \mathbb{R},$$
$$\mathrm{spec}(A) = \{a_0 - |\boldsymbol{a}|, a_0 + |\boldsymbol{a}|\}.$$

(5.4)

The Pauli matrices here are self-adjoint combinations of these matrix units. They initially represent the population difference and coherence components in this arithmetic observation cell; their specific physical identification is determined by the carrier and readout.

To turn a spectral decomposition into actual records, let $J_D : \mathbb{C}^2 \to \mathcal{H}$ embed the observation cell. Decompose a self-adjoint A on this cell as in (5.1), embed its spectral projections into the full space, and retain the output outside the cell:

$$Q_\kappa = J_D E_\kappa J_D^\dagger,$$
$$Q_\perp = I - J_D J_D^\dagger, \qquad \sum_r Q_r = I.$$

(5.5)

The index r includes all spectral outcomes and the complementary outcome. Choose a detector ready state $|g_D\rangle$ and record states $|r_D\rangle$ that together form an orthonormal set, using the original spectral instrument's pointer with zero bare energy and trivial group action. In the interaction picture, the recording exchange is

$$B_D = \sum_r Q_r \otimes \left(|r_D\rangle\langle g_D| - |g_D\rangle\langle r_D| \right),$$
$$W_D = e^{\pi B_D/2},$$
$$W_D(|\psi\rangle \otimes |g_D\rangle) = \sum_r Q_r |\psi\rangle \otimes |r_D\rangle.$$

(5.6)

The orthogonal projections correlate system components with their respective orthogonal records, while coherence between components remains in the full output. This readout is implemented over a finite duration t_D by the self-adjoint interaction $H_D^{\mathrm{int}} = i\pi\hbar B_D/(2t_D)$; the detection time belongs to the complete physical process.[8]

The conjugate address pair lies within a single energy shell, so these spectral projections commute with the free shell energy, and the complete detection operation is $(U_f(t_D) \otimes I_D) W_D$. For spectral readouts requiring a directional reference, the original construction also includes a regular group reference with zero bare energy in the apparatus. Write $G = G_{p,K}$, let R_g be the unit-group action on the system, and let a group element h act on the reference by $|g\rangle \mapsto |hg\rangle$. The lifted projections are

$$\tilde{Q}_r = \sum_{g \in G} (R_g Q_r R_g^\dagger) \otimes |g\rangle\langle g|.$$

(5.7)

On the reference state $|1_G\rangle$, this realizes the original Q_r; on the full space, it commutes with the joint group action on system and reference. The actual reference specifies the readout direction, and its degrees of freedom are retained with the apparatus. The spectral projections, record coupling, and conservation-compatible implementation thus belong to the same process.

The projections in the spectral decomposition therefore have an operational meaning: they specify the components distinguished by the apparatus, and the interaction correlates those components with records. The self-adjoint operator's eigenvalues assign real-valued readouts to these records. When a system is in a superposition of spectral components, the same apparatus acts on all of them without specifying an outcome in advance.

5.3 Observation of Quantum Coherence

The state read by the observation cell is $R_D = J_D^\dagger \rho J_D$. It is positive, and its trace is the cell's share of the full state. The record shares produced by the original spectral instrument satisfy

$$\begin{aligned} w_\kappa &= \mathrm{Tr}(R_D E_\kappa), \\ w_\perp &= 1 - \mathrm{Tr} R_D, \\ \textstyle\sum_\kappa a_\kappa w_\kappa &= \mathrm{Tr}(R_D A). \end{aligned}$$

(5.8)

The complementary outcome keeps the full output normalized. Conditional on selecting the cell, its state is $R_D/\mathrm{Tr}R_D$, with $\mathrm{Tr}R_D > 0$.

The same cell can read diagonal populations and off-diagonal coherence. For three separately configured spectral instruments, take

$$\begin{aligned} E_{i,\pm} &= \tfrac{I \pm \sigma_i}{2}, \\ w_{i,+} - w_{i,-} &= \mathrm{Tr}(R_D \sigma_i), \\ w_{i,+} + w_{i,-} &= \mathrm{Tr} R_D. \end{aligned}$$

(5.9)

The third direction reads the population difference between the two addresses, while the first two read the two real components of the off-diagonal coherence. Together with $\mathrm{Tr}R_D$, these readouts determine all of R_D. Relative phase in the complex amplitudes thus enters physical records.

Different spectral settings produce their respective outcomes separately; noncommuting quantities are not assigned a single joint sharp readout.

Complex coherence distinguishes matter patterns and carries their evolving relative phases. Thermal processing in the unified theory of matter retains reversible auxiliaries and thermal records. Through compensation, filtering, and return, the survival operation for each thermal record η becomes $\sqrt{r_\eta}\, I$ within the carrier. The total return share is $q_T = \sum_\eta r_\eta$, and the remaining outputs are retained as exits. During processing, the active dynamics is paused and bare free phases are compensated, so all components return with a common share. Let E_a, E_b be eigenvalues of the persistent generator h_X, and let t_H be the active duration of phase accumulation. The corresponding off-diagonal coherence is multiplied in each cycle by[9]

$$\lambda_{ab} = q_T e^{-i(E_a - E_b)t_H/\hbar}, \qquad 0 < q_T \leq 1. \tag{5.10}$$

Starting from nonzero initial coherence, two orthogonal spectral settings applied separately at the ends of repeated trials with identical preparations determine the evolved complex coherence. Continuous phase tracking, or alias-free sampling within a known frequency band, gives the unwrapped phase and its frequency. For a complete cycle period $\tau = t_H + T_{\text{proc}}$, the frequency relative to the cycle's physical time is

$$\omega_{ab}^{\text{obs}} = \frac{t_H}{\tau}\frac{E_a - E_b}{\hbar}. \tag{5.11}$$

The duration T_{proc} includes repair, thermal processing, filtering and return, compensation of free phases, and waiting in each cycle; final detection time is counted separately. The readable quantity in (5.11) is the phase difference between two components. The overall phase of a single pure state cancels in its pure-state projection.

The joint observable O_{pq} in Chapter 3 likewise shows that the scope of observation is determined by the actual interaction. Two states can have identical restrictions to all single channels yet be distinguished by a self-adjoint observable connecting two channels.[10] Local and interchannel coherences each have corresponding observable structures, which the apparatus must actually access.

The self-adjoint part of the arithmetic observable algebra supplies real spectra, and detector exchanges correlate spectral components with orthogonal records. Different apparatus settings can therefore read populations, off-diagonal coherence, and persistent phases; the required references remain in the joint process with the apparatus. These records already carry shares determined by the dynamics. Interpreting those shares as event probabilities then gives the linear pairing with a state the meaning of a statistical average of observations.

6 The Born-Probability Axiom

6.1 Mathematical Formulation of Event Probabilities

The Born-probability axiom specifies the statistical correspondence between states and observational outcomes. Born introduced the probabilistic interpretation of squared amplitude moduli in scattering theory; the Hilbert-space formulation expresses it as a pairing of states with spectral projections.[1,18] For the ideal spectral measurement (5.1), the probability of outcome a_κ is

$$\begin{aligned} p_\kappa &= \mathrm{Tr}(\rho E_\kappa), \\ p_\kappa &= \| E_\kappa \psi \|^2 \quad (\rho = P_\psi). \end{aligned}$$

(6.1)

For a rank-one spectral projection $E_\kappa = |\phi_\kappa\rangle\langle\phi_\kappa|$, this becomes the familiar $|\langle\phi_\kappa|\psi\rangle|^2$. A general apparatus can include finite efficiency, failure, and no response. Its outcomes are represented by positive effects E_r satisfying $\sum_r E_r = I$, with probabilities again given by $p_r = \mathrm{Tr}(\rho E_r)$.

In this framework, the effects arise from the recording process already implemented. Let η be the environment preparation, including the record registers in their ready states, $\mathbb{U}$ the full operation, and $\{\Pi_r^{\mathrm{rec}}\}$ the complete orthogonal family of actual record projections, including no-response outcomes. Define

$$\begin{aligned} \Omega^{\mathrm{out}}(\rho) &= \mathbb{U}(\rho \otimes \eta)\mathbb{U}^\dagger, \\ \mathcal{P}_r &= I \otimes \Pi_r^{\mathrm{rec}}, \\ \mathcal{I}_r(\rho) &= \mathrm{Tr}_E\left[\mathcal{P}_r \Omega^{\mathrm{out}}(\rho)\mathcal{P}_r\right]. \end{aligned}$$

(6.2)

These conditional operations connect record shares to the corresponding system changes and form a quantum instrument in the sense of Davies and Lewis.[3] Ozawa systematically studied the relation between measurement processes and completely positive instruments.[19] Here the original recording process has already specified the environment preparation and joint unitary. Expanding the partial trace in (6.2) gives their explicit operator representation:[8]

$$
\begin{aligned}
&J_{r\mu j} = \sqrt{\pi_j}\ \langle r, \mu | \mathbb{U} | j \rangle, \\
&\mathcal{J}_r(\rho) = \sum_{\mu,\, j} J_{r\mu j} \rho J_{r\mu j}^{\dagger}, \\
&E_r = \sum_{\mu,\, j} J_{r\mu j}^{\dagger} J_{r\mu j}, \qquad \sum_r E_r = I.
\end{aligned}
$$

(6.3)

Here $\eta = \sum_j \pi_j |j\rangle\langle j|$, and μ labels unresolved environment outputs within the same record. The pointer record r determines the event classification. Summing over unread indices within a record allows the same instrument component to have different operator representations.

The full operation has already determined the trace of each positive record component,

$$
\begin{aligned}
&w_r(\rho) = \mathrm{Tr}\mathcal{J}_r(\rho) = \mathrm{Tr}(\rho E_r), \\
&w_r \geq 0, \qquad \sum_r w_r = 1.
\end{aligned}
$$

(6.4)

These relations give the record shares supplied by the dynamics. A statistical assignment on the record branches gives them the meaning of event probabilities.

6.2 Probability Weights of Record Branches

Uniqueness of a probability rule depends on the consistency requirements imposed. Gleason derived the trace form from additive assignments to orthogonal projections in Hilbert spaces of dimension at least three; Busch obtained a version including dimension two by using consistent assignments on all positive effects.[4,5] We follow the argument in the original record theory. Its equal-amplitude exchanges and auxiliary refinements use Zurek's environment-assisted invariance approach and apply to the joint record branches already formed in the preceding section.[6,8]

Orthogonal records decompose the full pure state into joint components for the different outcomes. Let

$$
\begin{aligned}
&|\Psi\rangle = \sum_r a_r |\sigma_r\rangle, \\
&|\sigma_r\rangle = \frac{\mathcal{P}_r|\Psi\rangle}{a_r}, \qquad a_r = \| \mathcal{P}_r \Psi \| , \\
&\langle \sigma_r | \sigma_s \rangle = \delta_{rs}, \qquad \sum_r a_r^2 = 1,
\end{aligned}
$$

(6.5)

Normalized components are defined only for $a_r > 0$. Each σ_r contains the complete correlations between the system and recording apparatus for that outcome.

A statistical assignment must be consistent for the same physical outcome. Probabilities are nonnegative, add for mutually exclusive outcomes, and sum to one; zero components produce no events. Under a common unitary transformation of the joint state and readout, outcome probabilities transform covariantly. Unitary transformations confined to unread auxiliaries, addition of uncoupled auxiliaries, and copying or refinement of orthogonal records leave the probabilities of the original coarse outcomes unchanged. A mixed state and its purification give the same weights when the purification reference is unread, and continuous changes of state under the same readout give continuous probabilities.

Using these requirements, the state-space dynamics first determines probabilities for equal-amplitude outcomes by branch exchange, then extends them to general amplitudes by record refinement.[8]

First refine each outcome subspace into the one-dimensional direction of σ_r and its orthogonal complement. The latter has zero component in the current state, so this refinement leaves the original outcome probability unchanged. Next copy the existing orthogonal outcomes to an unread auxiliary, obtaining $\sum_r a_r|\sigma_r\rangle \otimes |\eta_r\rangle$, with orthogonal η_r. If two coefficients are equal, exchanging σ_r and σ_s and then exchanging their auxiliary labels leaves the full state unchanged. Covariance makes the first exchange permute outcome probabilities, while auxiliary consistency makes the second exchange leave the original probabilities unchanged. The two probabilities must therefore coincide. For N equal-amplitude components, normalization gives probability $1/N$ for each outcome. This copying applies only to record labels that are already orthogonal.

Next suppose the nonzero branches satisfy $a_r^2 = m_r/N$, where the m_r are positive integers with $\sum_r m_r = N$. Refine each auxiliary label η_r into m_r orthogonal label pairs in two auxiliaries, with coefficient $1/\sqrt{m_r}$ for each pair. The normalized images are mutually orthogonal, so this map extends to an auxiliary unitary. Include one auxiliary in the fine outcome space and retain the other as an unread orthogonal copy. Each fine branch then has coefficient

$$\frac{a_r}{\sqrt{m_r}} = \frac{1}{\sqrt{N}}.$$

(6.6)

The refined state has N equal-amplitude components, each with probability $1/N$. Recombining the mutually exclusive labels belonging to the same coarse outcome gives

$$p_r = \frac{m_r}{N} = a_r^2.$$

(6.7)

Every nonnegative normalized set of weights can be approximated by rational weights in the original outcome space. Continuity extends (6.7) to all normalized pure states. Zero components

receive zero probability, so the full result is $p_r(\Psi) = \| \mathcal{P}_r \Psi \|^2$. Auxiliary refinement tests the consistency of a single statistical rule; it does not change the outcome set already determined by the actual instrument.

For a general mixed state Ω, choose a purification $|\widehat{\Omega}\rangle$ on an unread auxiliary. The readout acts only on the original joint system, and purification consistency gives

$$\begin{aligned} p_r(\Omega) &= \| (\mathcal{P}_r \otimes I)\widehat{\Omega} \|^2 , \\ p_r(\Omega) &= \mathrm{Tr}(\mathcal{P}_r \Omega). \end{aligned}$$

(6.8)

Substituting $\Omega = \Omega^{\mathrm{out}}(\rho)$ yields

$$p_r(\rho) = \mathrm{Tr}\mathcal{J}_r(\rho) = w_r(\rho) = \mathrm{Tr}(\rho E_r).$$

(6.9)

This step identifies the record shares calculated from the dynamics as event probabilities. Squared-modulus weights are compatible with the orthogonal decomposition of Hilbert space and preserve the same statistical interpretation under outcome refinement, auxiliary transformations, and mixed-state representations.

For an ideal spectral instrument, the effects are E_κ, and (6.9) recovers (6.1). The self-adjoint pairing in Chapter 5 also becomes the average of actual outcomes: $\langle A \rangle = \sum_\kappa a_\kappa p_\kappa = \mathrm{Tr}(\rho A)$. Complex state amplitudes, orthogonal apparatus records, and observational statistics meet at this point.

6.3 Statistical Laws of Event Histories

An outcome also specifies the conditional state from which the subsequent process continues. For $p_r > 0$, the conditional state and the actual instrument component of the ideal spectral readout in Chapter 5 are, respectively,

$$\begin{aligned} \rho_r &= \frac{\mathcal{J}_r(\rho)}{p_r}, \\ \mathcal{J}_r(\rho) &= U_f(t_D) Q_r \rho Q_r U_f(t_D)^\dagger. \end{aligned}$$

(6.10)

In the interaction picture, the second expression reduces to $Q_r \rho Q_r$, whose normalization is the Lüders projection update.[20] Spectral outcomes within the cell lie in the same shell, so their free phase cancels. The complementary outcome can retain coherence between shells and must still include free evolution over the detection interval. For a general apparatus, the update remains

determined by $\mathcal{J}_r$: the same effect E_r can have different physical successors. The conditional state describes the subsequent system given a known record, while the complete joint state continues to retain all outputs and their correlations.

Repeated records of the original cycle form an ordered event history. With the same instrument, each fresh environment is independent of the system together with all earlier environments and records; old records are retained and no longer coupled. The resulting history $\boldsymbol{r} = (r_1, \ldots, r_N)$ satisfies[8]

$$\begin{aligned} &\mathcal{J}_{\boldsymbol{r}} = \mathcal{J}_{r_N} \circ \cdots \circ \mathcal{J}_{r_1}, \\ &p(\boldsymbol{r}) = \mathrm{Tr}\mathcal{J}_{\boldsymbol{r}}(\rho_0), \\ &\textstyle\sum_{\boldsymbol{r}} p(\boldsymbol{r}) = 1. \end{aligned}$$

(6.11)

When used auxiliaries re-enter feedback, the evolving state includes those auxiliaries and their correlations, and the history is calculated successively using the corresponding joint instrument. Cosmological event histories likewise carry forward the resources and correlations left by each interaction.[10]

Equation (6.11) also gives the conditional probability $p(s \mid r) = \mathrm{Tr}\mathcal{J}_s(\rho_r)$. An earlier event changes the successor state, so a later event is generally correlated with it; successive instrument actions on conditional states determine these correlations.

For independently repeated trials with the same preparation and instrument, let $\widehat{p}_r^{(L)}$ be the frequency of outcome r in L records. The original probability distribution gives

$$\begin{aligned} &\mathbb{E}\widehat{p}_r^{(L)} = p_r, \\ &\mathrm{Var}(\widehat{p}_r^{(L)}) = \frac{p_r(1-p_r)}{L}, \\ &\Pr\left(|\widehat{p}_r^{(L)} - p_r| \geq \delta\right) \leq \frac{p_r(1-p_r)}{L\delta^2}. \end{aligned}$$

(6.12)

The record frequency therefore converges in probability to the Born weight. For successive cycles of the same system, the state-space dynamics also controls correlations between records through a strict mixing bound for the complete cycle. In a stationary preparation, let $c > 0$ be the contraction constant already established for that cycle, r_* the stationary probability of a specified record, and $\bar{Z}_M$ its average count over M cycles. Then[8]

$$\Pr\left(|\bar{Z}_M - r_*| \geq \delta\right) \leq \frac{1 + 2/c}{M\delta^2}.$$

(6.13)

Through correlation decay, this result shows how successive events can also yield stable frequencies. Independent trials and successive cycles are supported by their respective preparations and dynamics. A count rate per unit physical time additionally requires division by the cycle duration, including all stages.

Specific arithmetic responses enter event statistics through the same rule. The unified theory of matter has implemented a complete spectral instrument on the $p = 7$ species space. Let S denote its success block on the species subspace. Its action is[9]

$$\begin{aligned}
&S\psi_\chi = R_\chi \psi_\chi \quad (\chi \neq 1),\\
&Se_0 = 0,\\
&R_\chi = \tfrac{L(1/2,\chi)}{\zeta(1/2)},\\
&E_{\text{succ}} = S^\dagger S, \qquad E_{\text{fail}} = I - S^\dagger S.
\end{aligned}$$

(6.14)

Here $L(s, \chi)$ is the Dirichlet L-function in the original spectral prescription, and the complete unitary implementation retains both success and failure. As a block of this unitary, S satisfies $S^\dagger S \leq I$. For the normalized nontrivial input direction ψ_χ, the actual success probability is $|R_\chi|^2$. The arithmetic spectrum determines an amplitude, and the Born rule turns it into an event probability for this instrument. Different χ label different input directions; these success probabilities do not require further normalization by summing over directions.

Joint evolution determines the record components, statistical consistency assigns them squared-modulus probabilities, and the actual instrument determines both the conditional state for each outcome and subsequent event histories. Amplitudes supplied by the arithmetic spectrum can therefore be compared with record statistics. Correlations between successive events are determined by the actual preparation, auxiliary use, and joint successors. States, interactions, and observational statistics connect within this process.

7 Conclusion

The evolution of matter and its observable manifestations have a common basis in the finite reduction of the Bost–Connes system and the joint interactions already constructed. Complex amplitude patterns describe matter, backgrounds and auxiliaries participate in its formation and maintenance, and measuring devices produce records through spectral coupling. The five axioms of quantum mechanics are thereby connected to a complete process encompassing states, interactions, and event statistics.

For a joint preparation containing all participants and their initial correlations, this connection is summarized by

$$\Omega^{\mathrm{out}} = \mathbb{U}\Omega^{\mathrm{in}}\mathbb{U}^{\dagger},$$
$$p_r = \mathrm{Tr}\left[(I \otimes \Pi_r^{\mathrm{rec}})\Omega^{\mathrm{out}} \right].$$

(7.1)

Here $\mathbb{U}$ includes formation, sustained dynamics, and record coupling, while Π_r^{rec} distinguishes the outcomes actually stored by the apparatus. The interaction determines the joint output and the weight of each record component; the statistical argument in Chapter 6 identifies these weights as event probabilities. Changes of state, the implementation of observation, and outcome statistics thus connect within the same complete process (Figure 7-1).

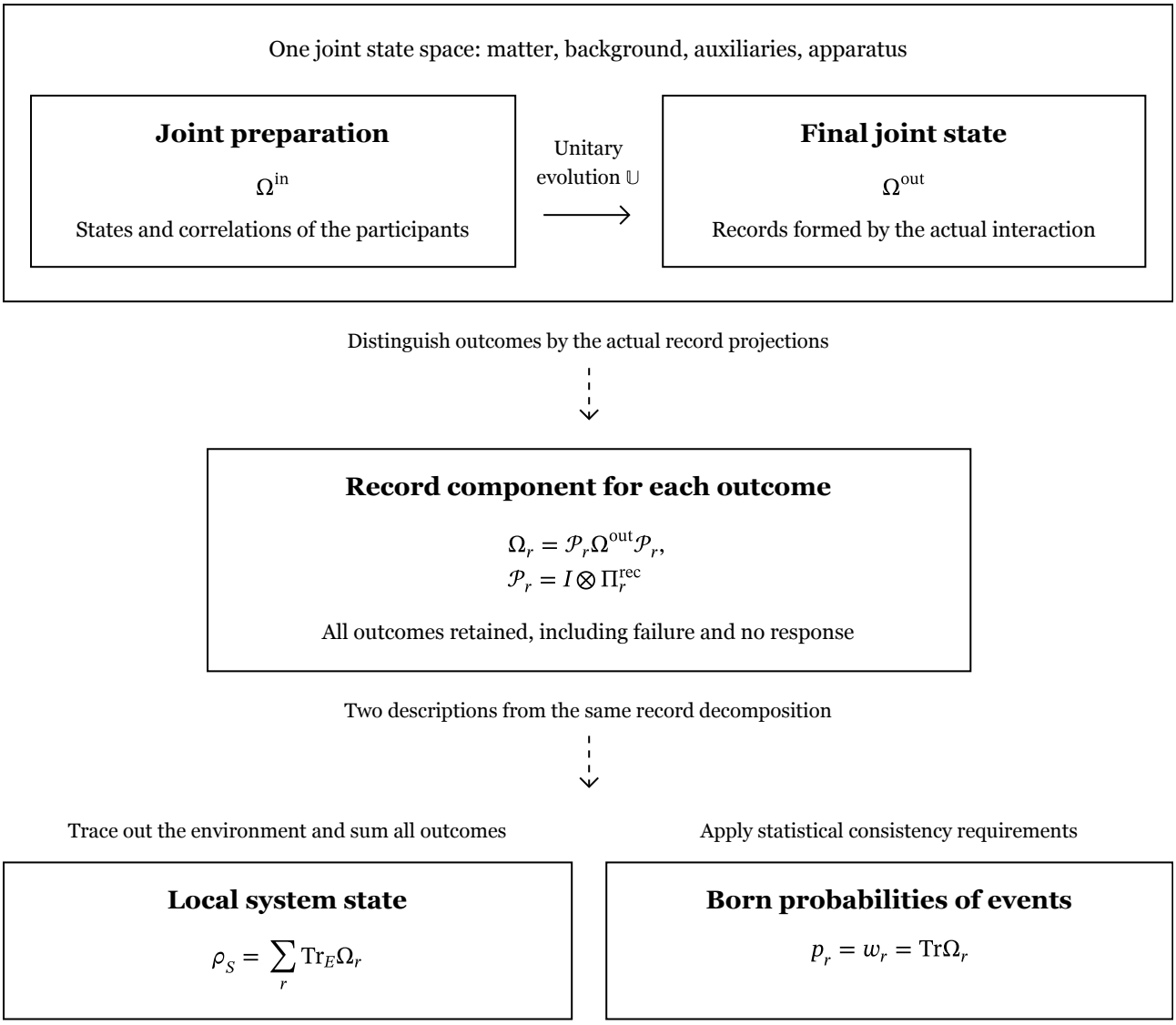


Figure 7-1. State evolution and event readout within a single joint process. The solid arrow represents actual unitary evolution; dashed arrows denote descriptive or statistical relations. The operator Ω_r is the diagonal block for record r in the final joint state; coherence between different records remains in that state. Tracing over the environment and summing all outcomes gives the system state, while the record shares receive a probabilistic interpretation under the statistical requirements of Chapter 6. The outcome set includes failure and no response.

Global coherence and local mixture consequently admit compatible descriptions. Formation exchanges establish system–environment correlations; local reduction retains system coherence according to overlaps of environment components, and subsequent exchanges continue to act on the original joint state. Conditional states given known records are obtained from the same output. The evolution of matter, apparatus backreaction, and correlations between events therefore belong to a single history.

The arithmetic structure further determines the specific physical content of this description. Character phases distinguish matter patterns, formation and maintenance determine their

subsequent evolution and persistent phases, and spectral responses supply the outputs of actual instruments. The general squared-modulus rule assigns probabilities, while the arithmetic dynamics determines the amplitudes; together they yield statistical predictions that can be compared with observed records.

This common construction expresses the compatibility of the five axioms: the complex linear structure of states accommodates superposition and composition, joint operations preserve its inner product, and internal spectral records connect distinctions in states to real-valued readouts and event statistics. Quantum mechanics thereby provides a unified form for describing matter patterns, interactions, and their observable manifestations within this framework.

References

[1] J. von Neumann. *Mathematische Grundlagen der Quantenmechanik*. Springer, Berlin (1932).

[2] K. Kraus. General state changes in quantum theory. *Annals of Physics* **64**, 311–335 (1971). doi: 10.1016/0003-4916(71)90108-4.

[3] E. B. Davies and J. T. Lewis. An operational approach to quantum probability. *Communications in Mathematical Physics* **17**, 239–260 (1970). doi: 10.1007/BF01647093.

[4] A. M. Gleason. Measures on the closed subspaces of a Hilbert space. *Journal of Mathematics and Mechanics* **6**, 885–893 (1957). doi: 10.1512/iumj.1957.6.56050.

[5] P. Busch. Quantum states and generalized observables: A simple proof of Gleason's theorem. *Physical Review Letters* **91**, 120403 (2003). doi: 10.1103/PhysRevLett.91.120403.

[6] W. H. Zurek. Probabilities from entanglement, Born's rule $p_k = |\psi_k|^2$ from envariance. *Physical Review A* **71**, 052105 (2005). doi: 10.1103/PhysRevA.71.052105.

[7] J.-B. Bost and A. Connes. Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory. *Selecta Mathematica* **1**, 411–457 (1995). doi: 10.1007/BF01589495.

[8] Shao-Wu Hu. *Dynamics of the Bost–Connes State Space*. Version 2. Zenodo (2026). doi: 10.5281/zenodo.23005935.

[9] Shao-Wu Hu. *The Unification of Matter, Forces, and Spacetime from the Bost–Connes System*. Version 8. Zenodo (2026). doi: 10.5281/zenodo.23554409.

[10] Shao-Wu Hu. *Cosmology and the Bost–Connes System*. Zenodo (2026). doi: 10.5281/zenodo.23639432.

[11] I. E. Segal. Irreducible representations of operator algebras. *Bulletin of the American Mathematical Society* **53**, 73–88 (1947). doi: 10.1090/S0002-9904-1947-08742-5.

[12] L. P. Hughston, R. Jozsa, and W. K. Wootters. A complete classification of quantum ensembles having a given density matrix. *Physics Letters A* **183**, 14–18 (1993). doi: 10.1016/0375-9601(93)90880-9.

[13] E. Schrödinger. Discussion of probability relations between separated systems. *Proceedings of the Cambridge Philosophical Society* **31**, 555–563 (1935). doi: 10.1017/S0305004100013554.

[14] R. F. Werner. Quantum states with Einstein-Podolsky-Rosen correlations admitting a hidden-variable model. *Physical Review A* **40**, 4277–4281 (1989). doi: 10.1103/PhysRevA.40.4277.

[15] W. F. Stinespring. Positive functions on C^*-algebras. *Proceedings of the American Mathematical Society* **6**, 211–216 (1955). doi: 10.1090/S0002-9939-1955-0069403-4.

[16] M.-D. Choi. Completely positive linear maps on complex matrices. *Linear Algebra and its Applications* **10**, 285–290 (1975). doi: 10.1016/0024-3795(75)90075-0.

[17] W. H. Zurek. Decoherence, einselection, and the quantum origins of the classical. *Reviews of Modern Physics* **75**, 715–775 (2003). doi: 10.1103/RevModPhys.75.715.

[18] M. Born. Zur Quantenmechanik der Stoßvorgänge. *Zeitschrift für Physik* **37**, 863–867 (1926). doi: 10.1007/BF01397477.

[19] M. Ozawa. Quantum measuring processes of continuous observables. *Journal of Mathematical Physics* **25**, 79–87 (1984). doi: 10.1063/1.526000.

[20] G. Lüders. Concerning the state-change due to the measurement process. English translation and discussion by K. A. Kirkpatrick. *Annalen der Physik* **518** (9), 663–670 (2006). doi: 10.1002/andp.20065180904.